

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of experience as a Software Developer and Data Engineer. Proven expertise in building scalable microservices and distributed data pipelines across diverse domains, leveraging a strong command of Java, Python, Spark, SQL, AWS, and Azure. Demonstrates a track record in CI/CD automation and graduate-level work on large-scale data platforms and privacy-preserving AI applications, including LLMs and vector databases.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Automated CI/CD pipelines with Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments for enhanced workflows.
- Modernized legacy systems by migrating to Log4j 2.x, improving monitoring and error handling to reduce issue resolution time by 20 percent.
- Spearheaded a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, utilizing Git, SVN, and JUnit testing for robust software development.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for automotive clients, demonstrating strong system design capabilities.
- Achieved a 90 percent system performance improvement through comprehensive SQL query optimization, database tuning, and legacy Java code modernization.
- Implemented CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates for efficient workflows.
- Mentored junior developers through code reviews, debugging sessions, and by promoting clean coding practices, fostering a culture of continuous learning.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for reliable software development.
- Supported release planning and sprint execution using JIRA within Agile development teams, demonstrating goal-oriented project execution.

PROJECTS

NYU Taxi Data Analysis Platform January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, showcasing expertise in distributed computing.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization capabilities.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and streamlined workflows.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent through system design principles.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, highlighting cloud technologies proficiency.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, leveraging LLMs and AI/ML skills.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, demonstrating advanced ML frameworks knowledge.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, exploring vector databases for potential future enhancements.

SKILLS

- Programming Languages: Proficient in Java, Python, and SQL for advanced data processing and software development.
- Frameworks & Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, and JUnit for

building scalable systems.

- Web Services: Skilled in developing and optimizing REST API and SOAP API for robust communication.
- Cloud Platforms: Experienced with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform, demonstrating strong cloud technologies knowledge.
- Databases: Proficient in PostgreSQL, Oracle Database, IBM DB2, and RDS, with an understanding of vector databases.
- Tools & Technologies: Adept with Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for infrastructure automation and workflows.
- Concepts: Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, and privacy-preserving LLM-based applications.
- Data Analysis & Visualization: Skilled in Tableau and QuickSight for reporting and decision-making.
- Soft Skills: Goal-Oriented, Decision-Making, and English proficiency, complemented by a strong track record in system design and distributed systems.